

1. D
2. B
3. B
4. A
5. A
6. A
7. B
8. C
9. A
10. C
11. B
12. C
13. B
14. C
15. D
16. D
17. B
18. C
19. D
20. C

21. Supervised learning is a type of machine learning algorithm that learn to predict new values based on past observation of a given labeled dataset.

22. Model M_1 is the result of ~~not~~ providing ~~enough~~ ^{too much} ~~the~~ training data Or not splitting the training and testing data set correctly result in a high accuracy for training while it performed poorly on new unseen data.

25. Major applications of data sciences are

- Recommendation system for streaming platform
- Image search for e-commerce

29. Data normalization is important and required for k-NN because k-NN is a type of lazy machine learning algorithm that doesn't need training. and All the data are mostly classified based on calculation of the distance between each point on the plane.

28. Data pre-processing is the process in a data science life cycle that involve cleaning data, handling missing data and/or transforming data so that the data is reliable for analysis and training.

27. To detect outlier in a data set, which is needed to analyse and produce reliable data we can calculate the z-score for each data row. which will determine if the data is an outlier depending on the result.

26. Distinguish between Structured and semi-structured data.

- Structured data are data that comes with an orderly manner in the form of rows and columns. For example database, spreadsheet.
- Semi-structured data are data that is structured but in an unordered manner. It can come in different format as well. For example JSON and XML.

